

Yougal Attri

AI/ML Engineer

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SUMMARY

AI/ML engineer with 2+ years of professional experience, including hands-on delivery of production-ready AI systems across computer vision and generative AI. Fine-tuned transformer-based neural networks (SAM, SegFormer) for satellite image segmentation and engineered end-to-end Python training and inference pipelines. Built a production-grade RAG chatbot over 317 RBI regulatory circulars with hybrid retrieval, cross-encoder re-ranking, and an LLM generation layer, scoring 0.96 answer relevancy on RAGAS.

TECHNICAL SKILLS

Machine Learning (ML): Machine Learning, Deep Learning, Neural Networks, Transformer Architectures, Model Fine-Tuning, Transfer Learning, Model Training, Model Evaluation & Optimization

Generative AI: Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Chatbot Development, Prompt Engineering, Embeddings, Semantic & Hybrid Search, Vector Databases (Qdrant)

Computer Vision: Image Processing, Image Segmentation, Geospatial & Satellite Imagery, OpenCV

AI Pipelines & Deployment: On-Prem and Cloud AI Deployment, Production-Grade AI Pipelines, End-to-End ML Workflows, Cloud AI Services (Google Cloud Platform), Docker, FastAPI, RESTful APIs, Scalable Architecture

Frameworks & Languages: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Pandas, NumPy, Python, SQL, JavaScript, C/C++, MongoDB, Git/GitHub

Core CS & Collaboration: Data Structures & Algorithms, Object-Oriented Programming (OOP), Cross-Functional Collaboration, Stakeholder Communication, Technical Documentation

PROJECTS

RBI Circular Intelligence System — Hybrid RAG Pipeline *github.com/Yougal17/rbi-rag*

- Built an end-to-end Retrieval-Augmented Generation application in Python over 317 RBI regulatory circulars (2022–2024, ~2.75M words), covering scraping, parsing, chunking, retrieval, and generation, with every answer citing its source circular, date, and URL.
- Engineered a custom Playwright scraper handling ASP.NET token-based filtering and Cloudflare-protected PDF delivery, and designed a hierarchical chunking strategy (120-word child / 600-word parent) producing 4,368 chunks indexed in a self-hosted on-prem Qdrant vector database.
- Implemented a 4-layer hybrid retrieval pipeline — dense vector search, BM25 over a 9,747-term vocabulary, Reciprocal Rank Fusion, and cross-encoder re-ranking — achieving 0.5–3.5s retrieval and 4–9s end-to-end latency, with a Gemini 2.5 Flash generation layer enforcing citations and hallucination guarding.
- Evaluated and tuned the pipeline with RAGAS across 30 test questions spanning 12 regulatory categories: 0.96 answer relevancy, 0.91 context precision, 0.775 overall average.

PROFESSIONAL EXPERIENCE

Executive — Network Operations | Spectra, Gurgaon, India *Aug 2026 – Present*

Fault Repair Service Engineer | Nextel Communications, Gurgaon, India *Jul 2025 – Jul 2026*

- Handled SLA-driven fault resolution for enterprise and retail customers, and maintained Excel-based tracking and reporting on fault trends, resolution times, and recurring failure causes for weekly operational review.

AI/ML Intern, then Full Stack Developer Intern | Panacea Infosec, New Delhi, India *Jun 2024 – Apr 2025*

- Fine-tuned SAM and SegFormer transformer-based neural networks on geospatial satellite imagery for image segmentation, improving performance roughly 70% over baseline through transfer learning and loss-function tuning for class imbalance.
- Engineered scalable, production-grade Python training and inference pipelines (PyTorch, TensorFlow, FastAPI), moving deep learning models from experimentation into production deployment.
- Built RESTful APIs with Node.js/Express.js and JWT-based authentication for a cybersecurity audit web application, collaborating with cross-functional teams to translate stakeholder requirements into technical solutions.

Data Science Intern | Sabudh Foundation *Jan 2024 – Jun 2024*

- Designed supervised and unsupervised machine learning models on social-impact datasets, building a classification pipeline that reached 95% accuracy.
- Streamlined data preparation and analysis workflows with Pandas and scikit-learn, cutting analysis turnaround time by roughly 30%.

PUBLICATION

“Blockchain in Supply Chain Management Industry: A New Era in Manufacturing Industry” — Book chapter, *Intelligent Manufacturing: Exploring AI, Blockchain, and Smart Technologies in Industry 4.0*, CRC Press (Taylor & Francis), March 2025.

EDUCATION

B.Tech (Honors), Computer Science & Engineering — Data Science Specialization *2020 – 2024*
Akala University, Bathinda, India | CGPA: 8.36 / 10